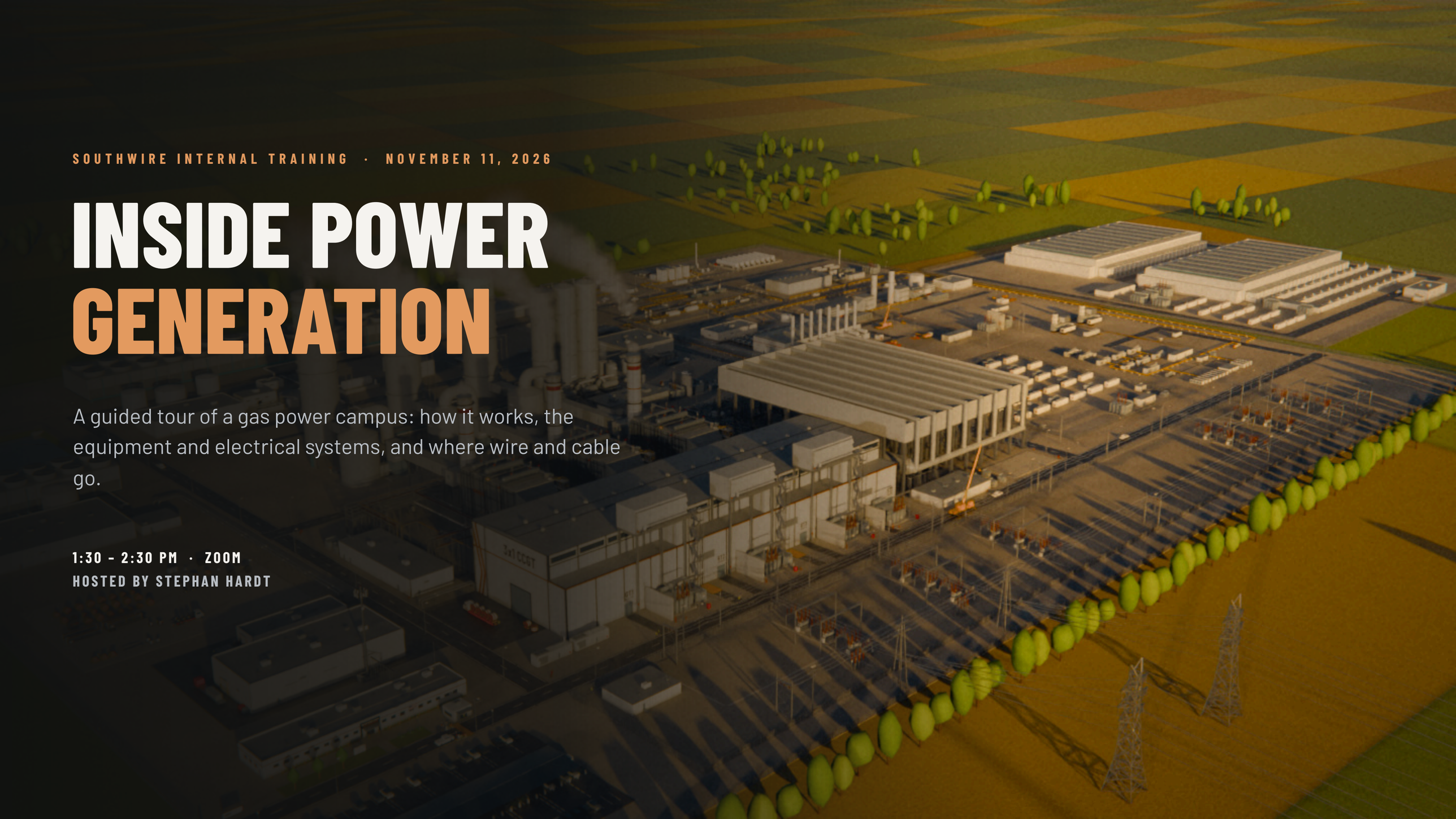
SOUTHWIRE INTERNAL TRAINING · NOVEMBER 11, 2026

INSIDE POWER GENERATION

A guided tour of a gas power campus: how it works, the equipment and electrical systems, and where wire and cable go.

1:30 - 2:30 PM · ZOOM

HOSTED BY STEPHAN HARDT



TODAY · 60 MINUTES

THE HOUR

1:30	**WELCOME** why this matters	3 MIN
1:33	**FUNDAMENTALS AND MARKET** why gas, how a combined cycle works, the campus	10 MIN
1:43	**THE 3D TOUR** seven stops: what it does, its electrical system, its cable	30 MIN
2:13	**WIRE AND CABLE TAKEAWAYS** the whole plant in one number, products, pain points	9 MIN
2:22	**Q&A** questions from the chat	8 MIN

No engineering background needed. At every stop you get three things: **what it does**, its **electrical system**, and **the cable**.

Questions in the chat; we answer them at the end.

stephan.hardt@southwire.com

WHY GAS, WHY NOW

Message: new gas build, modular and behind-the-meter generation, and upgrades to the existing fleet all need cable.

01

DEMAND IS GROWING AGAIN

Data centres and AI, manufacturing coming back, electrification of everything.

MARKET FIGURE: LOAD GROWTH / DATA-CENTRE DEMAND, FROM SOUTHWIRE MARKET INTELLIGENCE

02

FIRM POWER

Gas plants run when they are needed, day or night, and back up wind and solar.

MARKET FIGURE: NEW GAS CAPACITY PLANNED / UNDER CONSTRUCTION

03

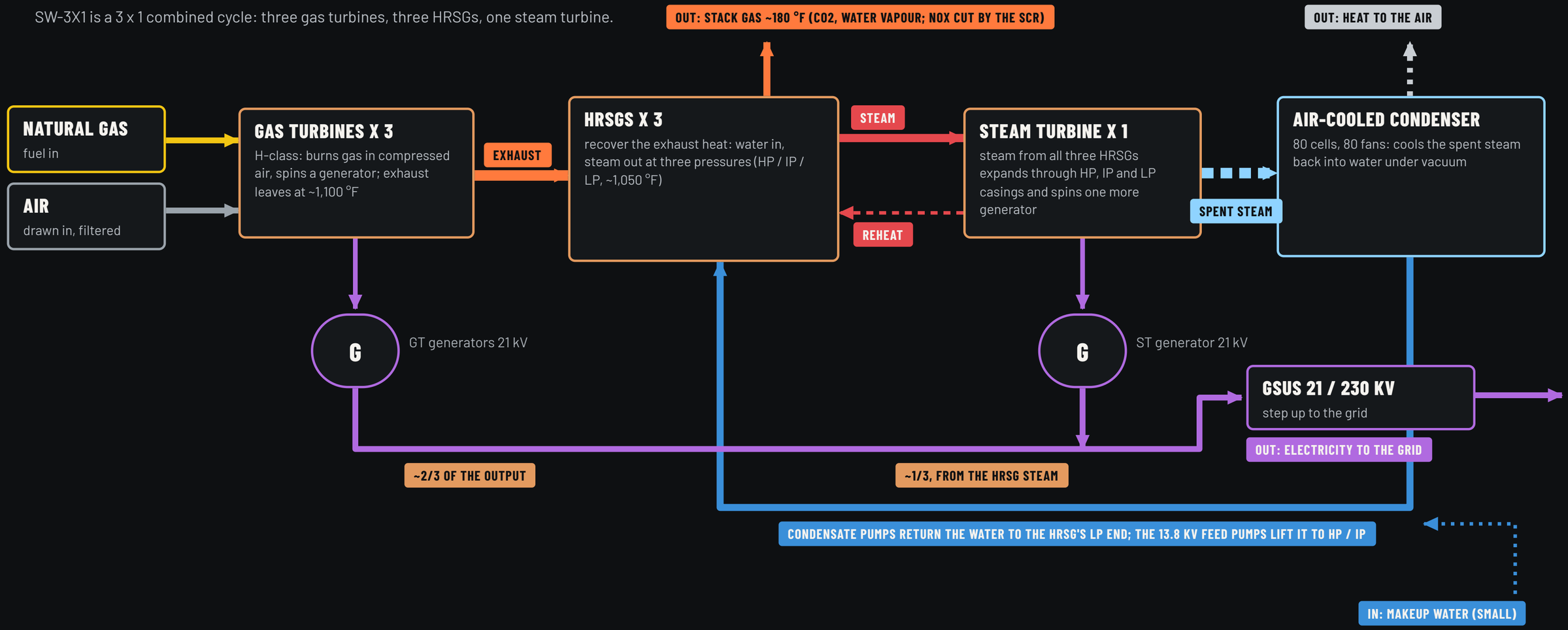
THREE WAYS IT BUYS CABLE

New gas plants, modular and behind-the-meter power at the customer, and upgrades of the existing fleet.

MARKET FIGURE: SIZE OF THE POWER-GENERATION CABLE MARKET FOR SOUTHWIRE

WHAT GOES IN, WHAT COMES OUT

SW-3X1 is a 3 x 1 combined cycle: three gas turbines, three HRSGs, one steam turbine.



IS IT STILL STEAM?

Leaving the turbine it is spent steam: ~100-150 °F, below atmospheric pressure, part water already. The air-cooled condenser turns it back into water.

A CLOSED LOOP

The same water goes round and round: pump, HRSG, turbine, condenser. Only a little makeup water replaces blowdown losses.

WHERE THE POWER COMES FROM

Fuel is burned only in the gas turbines (and duct burners). The steam turbine adds about half again on top, from heat that would otherwise go up the stack.

THE CABLE ON THE LOOP

21 kV isolated-phase bus to the GSUs; MV-105 to the feed pumps; VFD cable to 80 condenser fans; instruments on every pump, valve and level.

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FUNDAMENTALS · THE BASICS

HOW AN HRSG WORKS

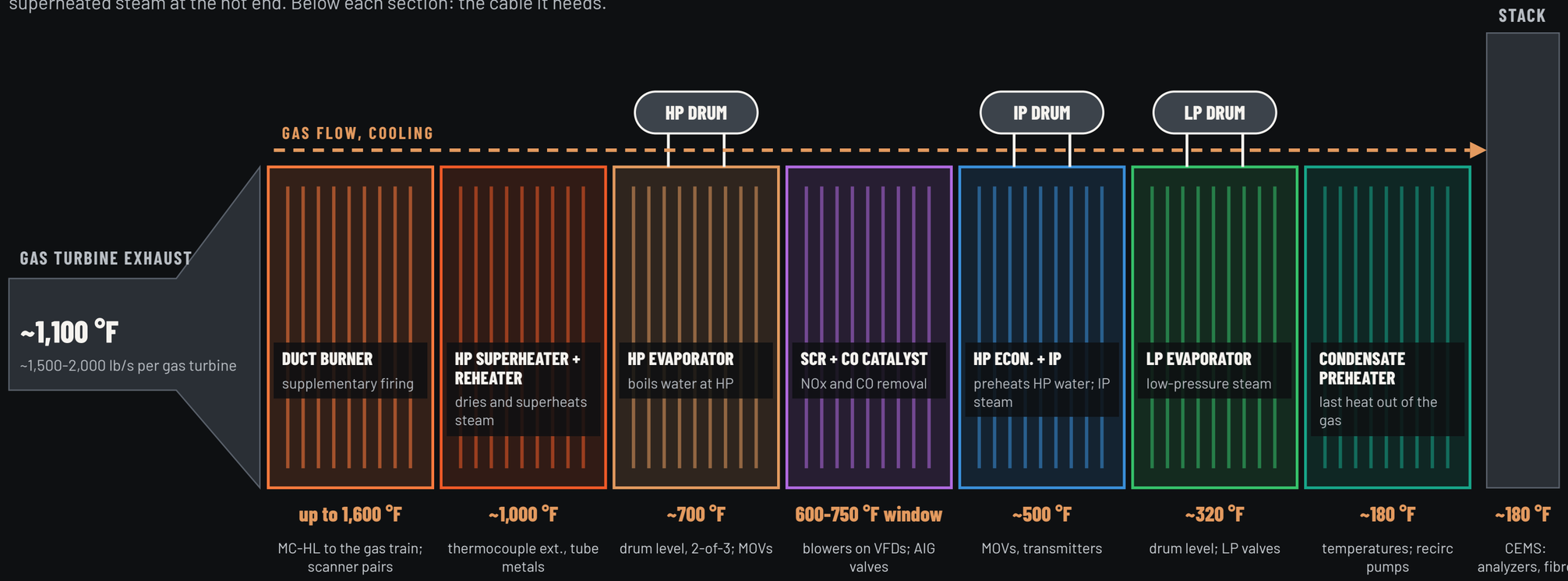
The exhaust crosses rows of finned tubes, hottest sections first; water enters at the cold end and leaves as superheated steam at the hot end. Below each section: the cable it needs.

WHAT IT IS

- **Without it:** ~60 % of the fuel's energy goes up the stack as ~1,100 °F exhaust (~40 % efficient).
- **With it:** a boiler heated by that exhaust; its steam drives the steam turbine.
- **The payoff:** ~1/3 more power from the same fuel (~60 %+ efficient).

FUN FACTS

- Fuel, CO2 and cost per MWh drop by about a third.
- The NOx catalyst (SCR) lives **inside** it, where the gas is just the right temperature.
- Duct burners can fire extra gas for peak demand.
- No HRSG: a gas-turbine plant. With one: combined cycle.



Steam to the steam turbine, typical: HP ~1,800-2,400 psig, ~1,050 °F · reheat / IP ~400-600 psig, ~1,050 °F · LP ~50-80 psig

~1/3
of a combined-cycle plant's output comes from HRSG steam, with no extra fuel (duct firing aside)

~60 %+
combined-cycle efficiency, against ~40 % for the gas turbine alone

3 x 3
three pressures, three drums; each drum level measured three ways, typically voted 2-out-of-3

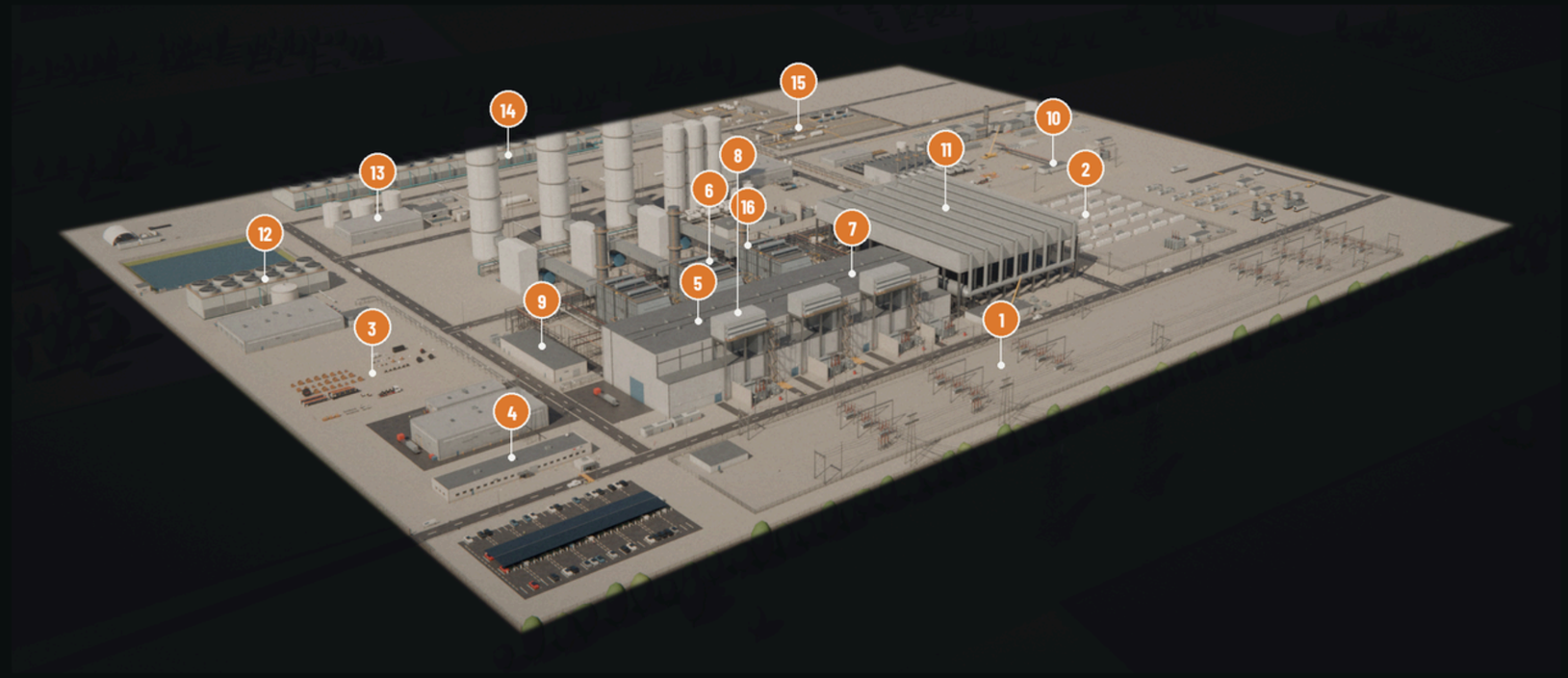
~1 hour
a hot start: about an hour: every start cycles the tubes, which is why tube-metal thermocouples matter

Typical three-pressure reheat HRSG with SCR behind an H-class gas turbine; section order, temperatures and pressures vary by design.

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THE PLANT IS THE MODEL

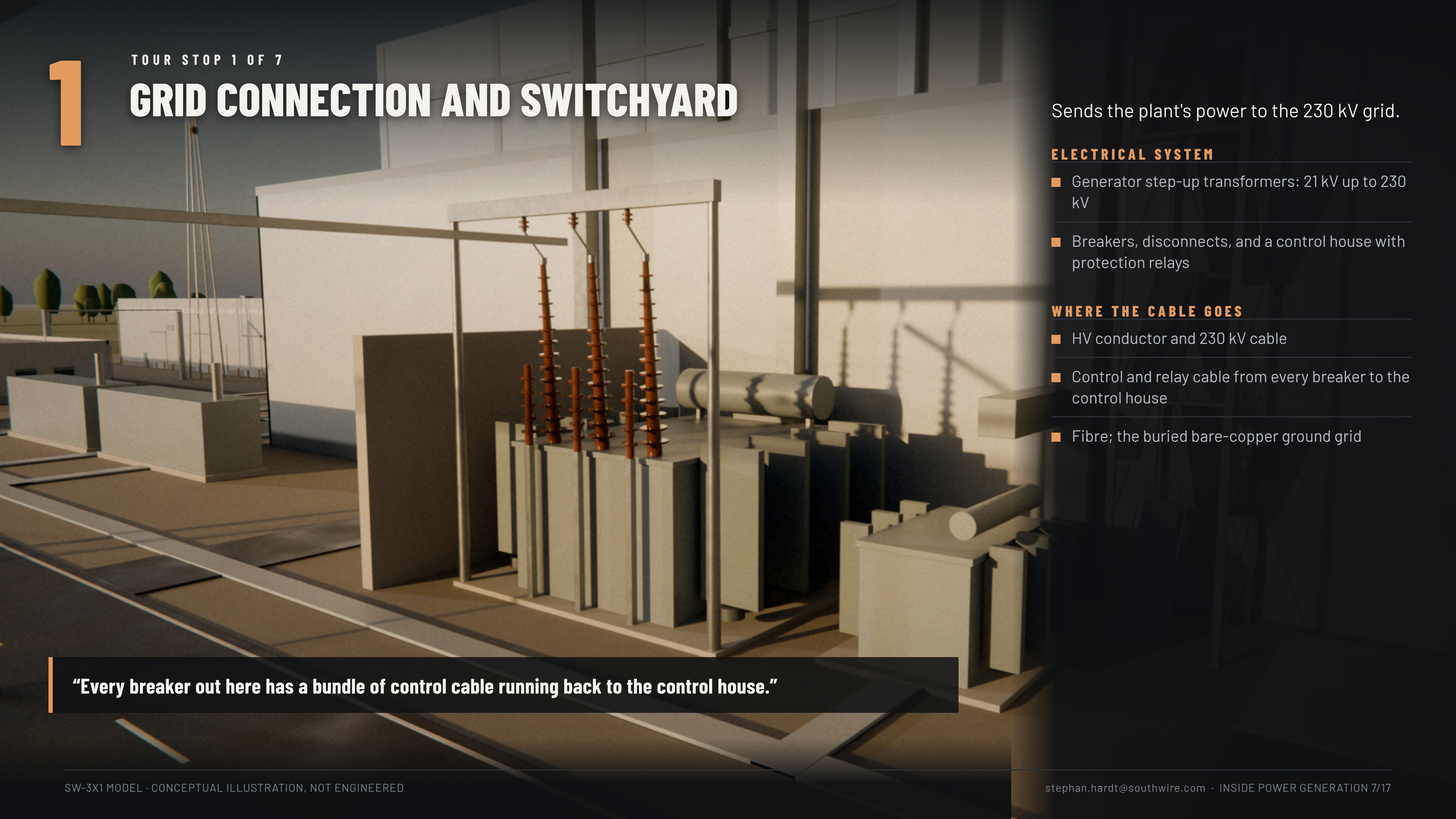
16 ZONES, ONE PLANT-WIDE CABLE SCOPE



01 Switchyard, GSU & grid tie
02 BESS yard
03 Reel staging & prefab
04 Admin & control room
05 E-house
06 HRSG trains & stacks
07 Turbine hall
08 GT air inlets
09 MCC, VFD & UPS room
10 Modular power yard
11 Air-cooled condenser
12 Chillers & cooling towers
13 Water treatment
14 Carbon capture (CCS)
15 Gas metering
16 Cable corridor & MV

PLANT · EQUIPMENT · CABLE · DELIVERY

SK-3X1 3x1 combined cycle campus · rendered from the plant model · conceptual, not engineered



TOUR STOP 1 OF 7

1 GRID CONNECTION AND SWITCHYARD

Sends the plant's power to the 230 kV grid.

ELECTRICAL SYSTEM

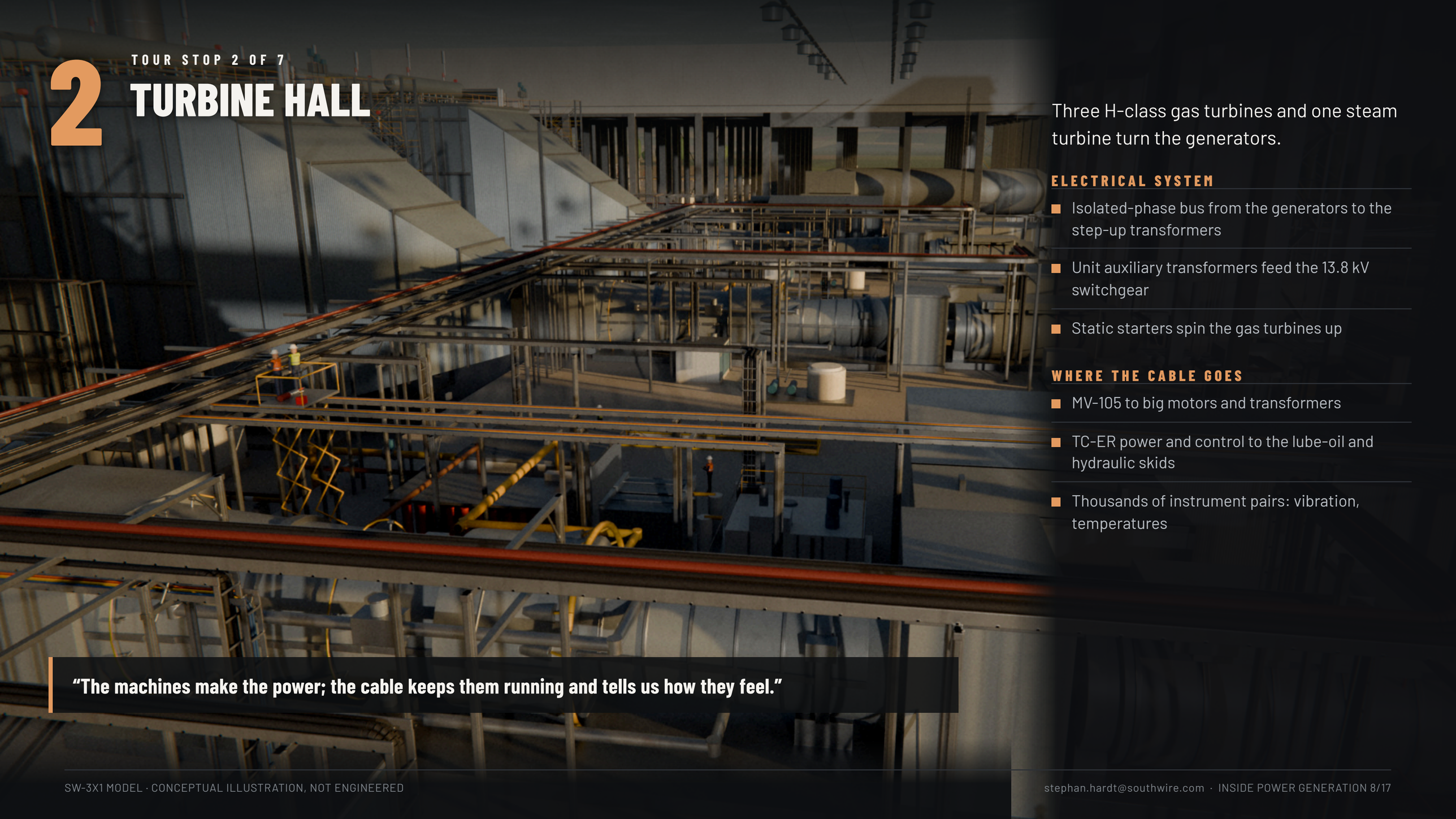
- Generator step-up transformers: 21 kV up to 230 kV
- Breakers, disconnects, and a control house with protection relays

WHERE THE CABLE GOES

- HV conductor and 230 kV cable
- Control and relay cable from every breaker to the control house
- Fibre; the buried bare-copper ground grid

> **“Every breaker out here has a bundle of control cable running back to the control house.”**

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TOUR STOP 2 OF 7

2 TURBINE HALL

Three H-class gas turbines and one steam turbine turn the generators.

ELECTRICAL SYSTEM

- Isolated-phase bus from the generators to the step-up transformers
- Unit auxiliary transformers feed the 13.8 kV switchgear
- Static starters spin the gas turbines up

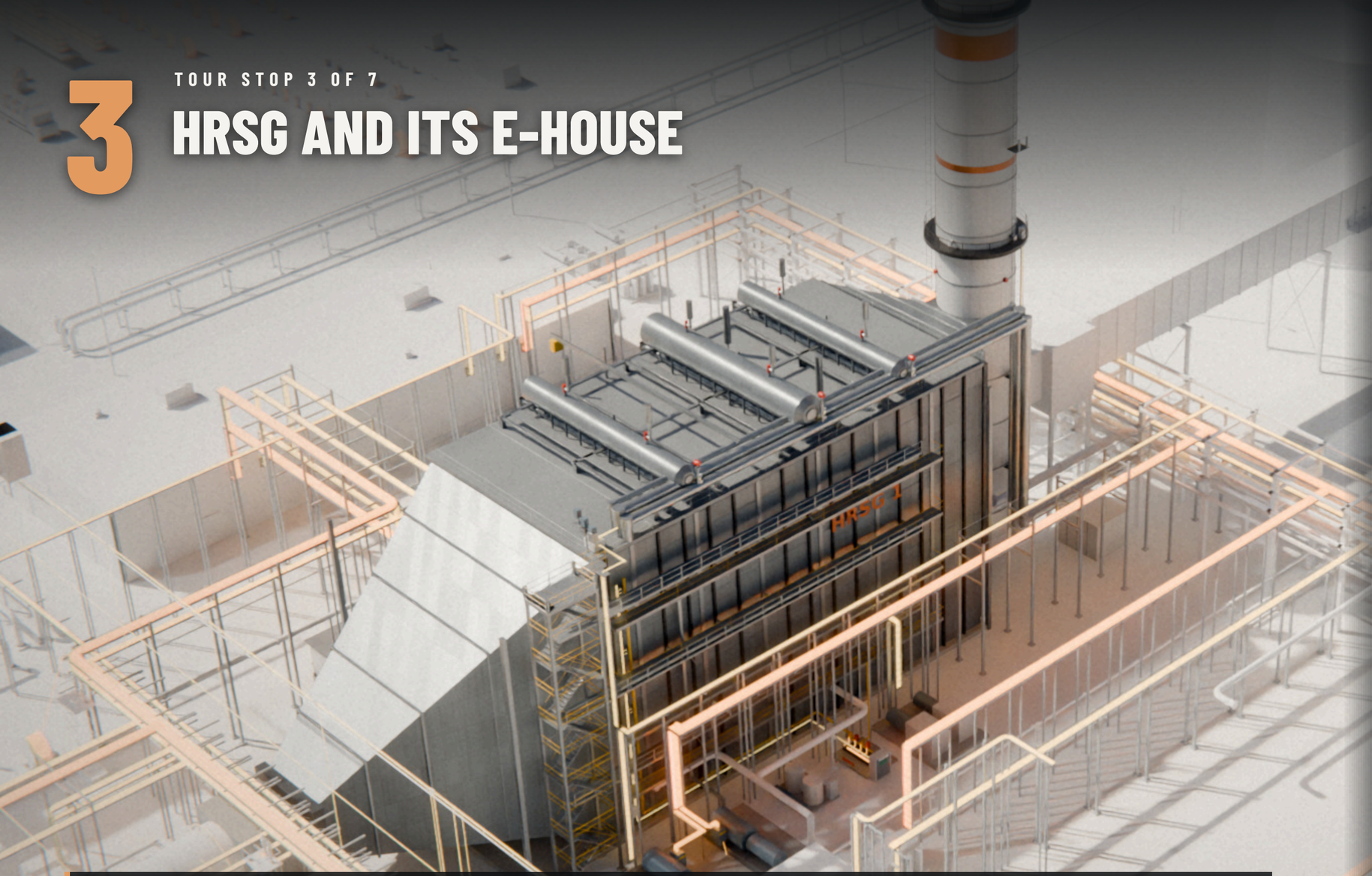
WHERE THE CABLE GOES

- MV-105 to big motors and transformers
- TC-ER power and control to the lube-oil and hydraulic skids
- Thousands of instrument pairs: vibration, temperatures

“The machines make the power; the cable keeps them running and tells us how they feel.”

TOUR STOP 3 OF 7

3 HRSG AND ITS E-HOUSE



Turns turbine exhaust heat into steam; the SCR inside cuts NOx.

ELECTRICAL SYSTEM

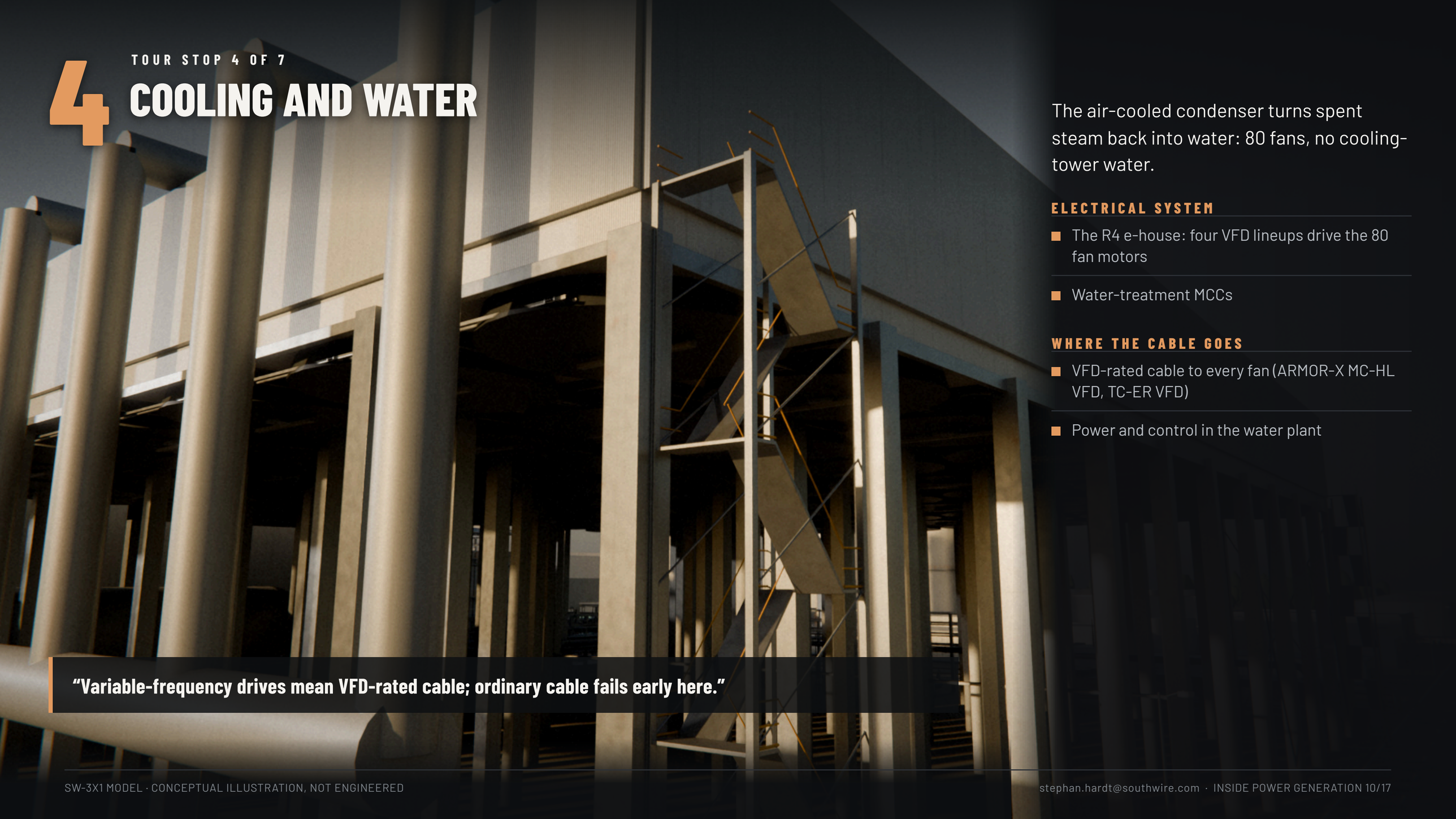
- Two 13.8 kV feed pumps
- A transformer feeds the e-house: 480 V switchgear, MCCs, drives, DCS
- CEMS shelter at the stack

WHERE THE CABLE GOES

- One HRSG train: 707 cables, about 157,000 ft
- About three quarters are instrument and thermocouple pairs
- ARMOR-X MC-HL at the duct-burner gas skid

“One boiler train is about 30 miles of cable, and most of it is small instrument cable.”

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED



TOUR STOP 4 OF 7

4 COOLING AND WATER

The air-cooled condenser turns spent steam back into water: 80 fans, no cooling-tower water.

ELECTRICAL SYSTEM

- The R4 e-house: four VFD lineups drive the 80 fan motors
- Water-treatment MCCs

WHERE THE CABLE GOES

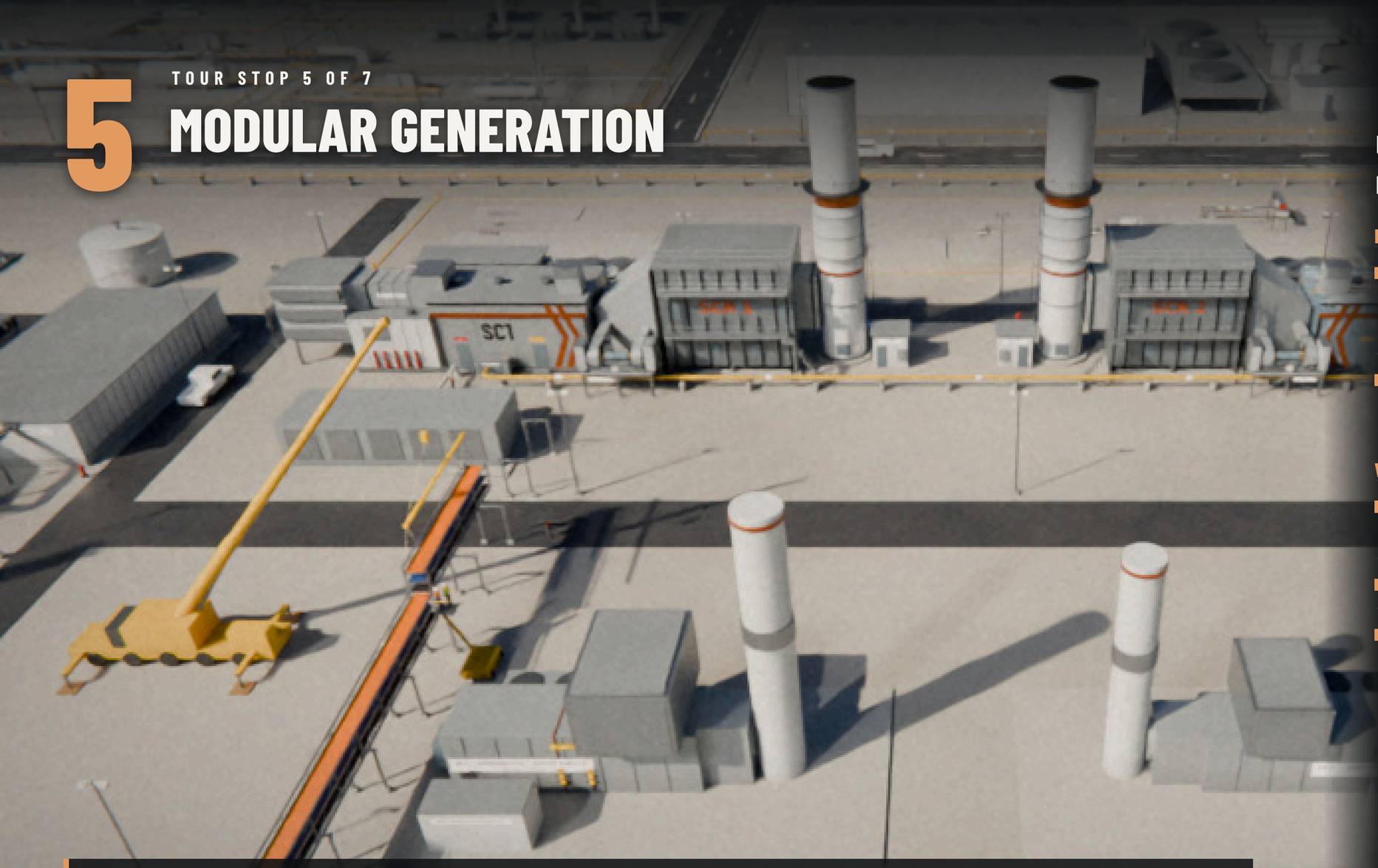
- VFD-rated cable to every fan (ARMOR-X MC-HL VFD, TC-ER VFD)
- Power and control in the water plant

“Variable-frequency drives mean VFD-rated cable; ordinary cable fails early here.”

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TOUR STOP 5 OF 7

5 MODULAR GENERATION



Fast-start, flexible power beside the big plant.

ELECTRICAL SYSTEM

- Gas engines in a hall, aeroderivative and industrial gas-turbine packages, containers, trailers, fuel cells, microturbines
- Each unit on the 13.8 kV collector; inverters for the battery and fuel cells

WHERE THE CABLE GOES

- MV collection cable; DC cable at the battery and fuel cells
- Factory pre-wired skids and plug-in modules
- Pre-terminated and cut-to-length cable

“Modular means factory pre-wiring: our opportunity for pre-terminated, cut-to-length cable.”

TOUR STOP 6 OF 7

6 BEHIND THE METER: THE DATA CENTRE



A data-centre campus powered directly by the plant's modular units, with the grid as backup.

ELECTRICAL SYSTEM

- 13.8 to 34.5 kV step-up transformers and double-bus switchgear
- A 40 MW battery absorbs load steps
- Two data halls fed 2N through pad-mount transformers and UPS

WHERE THE CABLE GOES

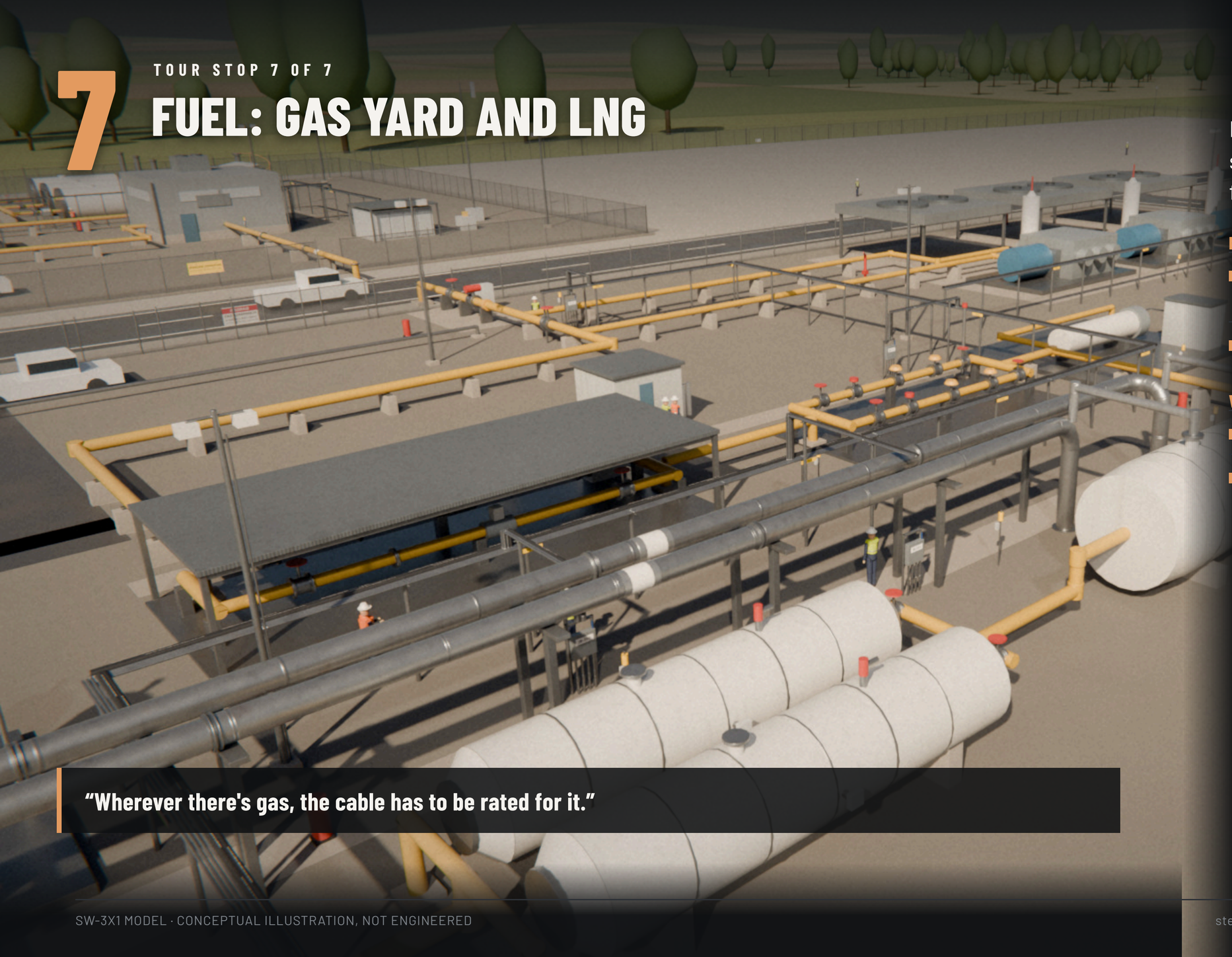
- 35 kV MV cable
- Large LV feeders to the UPS skids
- Fibre everywhere

> **“Data centres are why gas is growing again, and they are cable-dense on both the A and the B side.”**

SW-3X1 MODEL · CONCEPTUAL ILLUSTRATION, NOT ENGINEERED

TOUR STOP 7 OF 7

7 FUEL: GAS YARD AND LNG



Brings the fuel in and conditions it; an LNG satellite holds about three hours of backup fuel.

ELECTRICAL SYSTEM

- Metering and regulating station; the LNG e-house (13.8 kV / 480 V)
- Pumps, vaporizers, boil-off compressor

WHERE THE CABLE GOES

- ARMOR-X MC-HL for Class I Div 1 and 2
- TC-ER control, PLTC instrumentation and fire alarm on one strut rack

> **“Wherever there's gas, the cable has to be rated for it.”**

THE WHOLE PLANT IN ONE NUMBER

6.4M

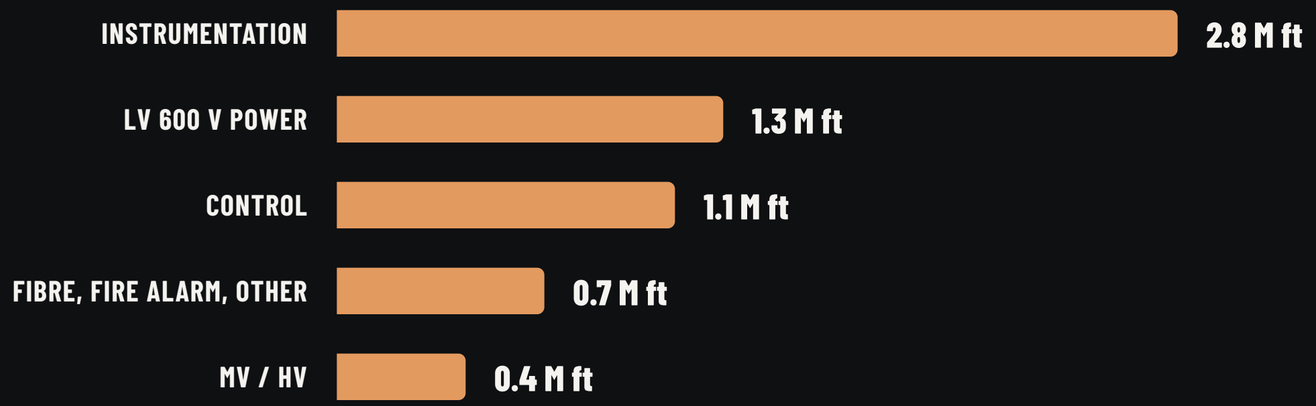
FEET OF CABLE · ABOUT 1,212 MILES

6,011
CABLE RUNS

21,217
CABLES

From the model's cable schedule: every circuit on the campus, from the 230 kV switchyard to the last thermocouple.

FOOTAGE BY CABLE TYPE



Most of the footage is small cable: instrument and control. MV is the high-value part, but it isn't the volume.

WHAT THE SPEC CALLS FOR

Every cable application on the HRSG, the requirements an owner's or EPC's cable specification writes for it, and the Southwire product built to meet them.

APPLICATION · WHERE	WHAT THE SPECIFICATION CALLS FOR	SOUTHWIRE
MV POWER, 15 KV FEED-PUMP MOTORS; THE T-R2A PRIMARY	1/C Cu, EPR insulation, **133 % level**, 90 °C, tape shield, PVC jacket. ICEA S-93-639, UL 1072 (MV-105). Terminations IEEE 48; acceptance test VLF (IEEE 400.2) / NETA ATS	**MV-105 15 kV, 220 mil NL-EPR 133 %, SIMpull · SPEC 46302**
480 V FEEDERS AND POWER T-R2A SECONDARY, LC-R2A TO MCCS, MOTORS AND MOVS	Cu XHHW-2, 90 °C wet / dry, sized at **125 %** of full load; voltage drop ≤ 3 %. ICEA S-95-658, UL 44 / UL 1277; 10+ parallel sets: compare with bus duct	**XHHW-2 1/C (SIMpull) · Type TC-ER 3/C + ground · SPEC 45252**
TRAY CABLE: POWER AND CONTROL EVERY TRAY BETWEEN THE E-HOUSE AND THE HRSG	Type TC-ER (UL 1277) so it may leave the tray as an exposed run to the equipment (NEC 336.10); sunlight resistant, direct burial; **IEEE 1202 / UL 1685** 70,000 Btu/h vertical-tray flame test	**Control TC-ER 14 AWG multiconductor · SPEC 45053**
HAZARDOUS AREA: GAS TRAIN, DUCT BURNER DUCT-BURNER FUEL SKID, BMS PANEL TO THE VALVES	Class I Div 2 wiring methods per NEC 501.10(B); MC-HL (UL 1569 + UL 2225) also qualifies for Div 1. Burner management per **NFPA 85**, duct burner UL 295, listed fittings	**ARMOR-X MC-HL, welded armor · SPEC 45352**
VARIABLE-FREQUENCY DRIVES SCR DILUTION-AIR BLOWERS, FD FANS	VFD-rated: symmetrical three-ground construction, overall shield or continuous armor for the common-mode current, XLPE insulation for reflected-wave peaks	**ARMOR-X MC-HL VFD · TC-ER VFD · SPEC 45352**
INSTRUMENTATION, 4-20 MA / HART DRUM LEVEL, PRESSURES, FLAME SCANNERS, ANALYZERS	**Individually shielded** twisted pairs + overall shield, drain wires; 300 V PLTC (UL 13, NEC 725) or ITC; ICEA S-73-532; segregated from power in tray	**Instrumentation PLTC, SPOS · SPEC 43303**
THERMOCOUPLE EXTENSION GAS-PATH, TUBE-METAL AND DUCT TEMPERATURES	Alloy matched to the thermocouple (type K = KX), ANSI MC96.1 / ASTM E230 colour code; **never copper**; shielded pairs; PLTC; 105 °C near the casing	**Thermocouple extension, PLTC multi-pair · SPEC 42345 / 42350**
FIRE ALARM, FIBRE, BUILDING WIRE F&G LOOPS, CEMS DATA, E-HOUSE LIGHTING	FPLR / FPLP (NEC 760, NFPA 72); single-mode fibre, armored in tray; THHN / THWN-2 in conduit (UL 83)	**SIMpull THHN / THWN-2 · fire-alarm and fibre cable**

CHECKED AGAINST A REAL PLANT

A recent US utility combined-cycle filing shows the same structure: **HRSG MCCs A / B** fed from 480 V switchgear through 2,500 / 3,300 kVA transformers off the auxiliary bus (6.9 kV there, 13.8 kV here).

DESIGN AND INSTALL

Raceway and cable design for generating stations: IEEE 422. Grounding: IEEE 80 / 665. Area classification: NEC 500, API RP 500. Field acceptance: NETA ATS.

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PAIN POINTS AND THE ANSWER

TECHNICAL PAIN POINTS

- **MOSTLY SIGNAL CABLE** ~74 % of an HRSG's cables are instrument or thermocouple pairs; bad shield grounding or routing next to drives and MV makes noisy drum-level and flame-scanner signals that can trip the unit
- **THERMOCOUPLE EXTENSION ERRORS** ~84 runs per train: wrong alloy, reversed polarity or copper substituted; found as a bad tube-metal reading after startup
- **HEAT AND SUN** Hot casings, roof penthouses, drum platforms: 90-105 °C ratings, sunlight-resistant jackets, derating
- **MOVEMENT AND CYCLING** The HRSG grows inches when hot and cycles daily: slack at module joints, vibration, termination fatigue
- **GAS IN THE BOILER** Duct burners and fuel skids make Class I Div 2 areas; NFPA 85 burner management; conversions add it
- **BIG PARALLEL FEEDERS** 3,007 A transformer secondary = 40 cables of 500 kcmil; equal lengths in congested tray
- **DRIVES** SCR blowers and fans on VFDs: bearing currents and voltage spikes without VFD-rated cable
- **SCOPE GAPS** Vendor junction boxes, EPC home runs: legs fall between the two (our own schedule missed the 480 V secondaries)
- **SCHEDULE AND LABOUR** Cable is on the critical path, above all in an outage; electricians are scarce; no room to stage reels

VALUE PROPOSITION

- **ONE SUPPLIER, WHOLE SCOPE** MV-105, TC-ER, ARMOR-X MC-HL and VFD, instrumentation, thermocouple extension, THHN, fire alarm: one PO, one spec-compliance package (IEEE 1202, UL 1277 / 2225 / 13)
- **FASTER PULLS: SIMPULL** Pre-lubricated jacket cuts pulling force and time; parallel conductors on one reel go in one pull
- **STAGED FOR THE OUTAGE** Cut-to-length reels labelled per pull, SIMpull Cradles for one-person payoff: ready before the trip
- **ONE CABLE FOR HAZARDOUS AREAS** ARMOR-X MC-HL is rated Div 2 and Div 1: fewer conduits and seals, unaffected by reclassification
- **THE RIGHT CONSTRUCTION** VFD-rated cable for drives, individually shielded pairs, matched-alloy colour-coded thermocouple extension
- **ENGINEERING UP FRONT** Ampacity, voltage drop, spec review; takeoff ready: ~707 cables, ~157,000 ft per train, ~472,000 ft for three
- **SUSTAINABILITY** Reel returns and scrap recycling for cable cut back in brownfield outages

"On an HRSG, cable is three-quarters instrumentation and fully on the critical path. One spec-compliant supply, staged and fast to pull: wired on time, reading right the first time."

stephan.hardt@southwire.com

THREE THINGS TO REMEMBER

TAKE AWAY

1 **GAS IS GROWING AGAIN**

driven by data centres and the need for firm power.

2 **MOST OF A PLANT'S CABLE IS SMALL**

instrument and control; the big MV and power cable is the smaller share.

3 **SOUTHWIRE COVERS THE WHOLE PLANT**

one supplier, staged on SIMpull reels, spec-compliant from MV down to the thermocouple.

Q&A

Questions from the chat. The recording and the 3D viewer link will be shared after the session.

Stephan Hardt
stephan.hardt@southwire.com